

Javier Fernandez-Marques

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SUMMARY

I am the lead research scientist at Flower Labs. We are building Flower, the go-to framework for all-things Federated AI. I am also a visiting researcher to the Machine Learning Systems Group at the University of Cambridge, where I collaborate with PhD students, and supervise Part-II (undergraduate) and MPhil (masters) dissertation projects.

Previously, I was a research scientist at Samsung AI Cambridge (UK) where investigated new hardware-aware optimizations for deep learning and I contributed towards areas such as Federated Learning, Approximate Computing and NAS. In 2021, I got my DPhil (PhD) in Computer Science from the University of Oxford.

WORK EXPERIENCE

FLOWER LABS

Lead Research Scientist

Cambridge, UK

June 2023 - now

- Building the tools for decentralized AI.

UNIVERSITY OF CAMBRIDGE

Visiting Researcher

Cambridge, UK

June 2022 - now

- I'm associated with the Machine Learning Systems Group, where I supervise Part-II (undergraduate) and Part-III/MPhil (masters) dissertation projects.

SAMSUNG AI

Research Scientist

Cambridge, UK

July 2021 - June 2023

- I design and implement new methods to accelerate ML workloads. Working at the intersection of neural architecture search, approximate computing and federated learning.

ARM

ML Research Intern

Cambridge, UK

May 2019 - Sept. 2019

- Developed a novel approach of using Winograd convolutions, enabling the deployment of INT8 models with marginal accuracy drops, no impact on model size and, up to 2× speedup compared to optimized INT8 *im2row* on Arm Cortex-A73 and A53 mobile CPUs.
- Designed a framework that transforms a given model (a CNN) into its Winograd-aware counterpart. This framework jointly optimizes accuracy and latency via NAS.

HOLITION

Computer Vision Engineer

London, UK

Oct. 2016 - Oct. 2017

- Lead computer vision and machine learning engineer developing AR applications for the fashion retail sector. Research and implementation of low-level computer vision techniques suitable for real-time execution on hand-held devices (smartphones, tablets)
- Maintained and enhanced our in-house lightweight face tracking algorithm.
- Worked on full application cycle: design, prototyping, implementation and optimisation.

HAWK-EYE INNOVATIONS (SONY)

Software Engineer

Basingstoke, UK

Feb. 2016 - Oct. 2016

- Research and implementation of low-level image processing (demosaicing and colour correction) and multi-camera calibration and player-tracking algorithms. Improved UI and UX of a camera calibration software used in hundred of sport events through the year.

FRAUNHOFER IIS

Research Intern

Erlangen, Germany

Summer 2014

- Investigated Literature on stereo vision and implemented a state of the art stereo-vision algorithm to extract the depth-maps from a scene using a multi-camera array.

EDUCATION

UNIVERSITY OF OXFORD

Oxford, UK

DPhil (PhD) in Computer Science

Oct. 2017 – 2021

- **Theme:** High Performing Deep Learning for Resource Constrained Platforms
- Supervised by Associate Prof. Nicholas D. Lane
- TA in Deep Neural Networks course for MSc students. (MT18, HL19, MT19)

QUEEN MARY, UNIVERSITY OF LONDON

London, UK

MSc in Computer Vision (Distinction)

Sept. 2014 - Sept. 2015

- **Thesis:** Pedestrian detection from varying viewing angles (89/100)
- Supervised by Professor Andrea Cavallaro

TECNUN, UNIVERSITY OF NAVARRA

San Sebastián, Spain

BSc in Telecommunications Engineering

Sept. 2010 - May 2014

- **Thesis:** Collagen Mesh Detection and Quantification in Reflection Microscopy (100/100)
- Co-supervised by Professor Carlos Ortiz-de-Solórzano and Associate Prof. Arrate Muñoz

SELECTED PUBLICATIONS

- ICML 2024 Royson Lee, **Javier Fernandez-Marques**, Shell Xu Hu, Da Li, Stefanos Laskaridis, Łukasz Dudziak, Timothy Hospedales, Ferenc Huszár, Nicholas D. Lane. *Recurrent Early Exits for Federated Learning with Heterogeneous Clients*. International Conference on Machine Learning (ICML), 2024.
- ACM TODAES 2023 Stylianos I. Venieris, **Javier Fernandez-Marques**, and Nicholas D. Lane. *Mitigating Memory Wall Effects in CNN Engines with On-the-Fly Weights Generation*. ACM Transactions on Design Automation of Electronic Systems (TODAES), 2023.
- INTERSPEECH 2022 Yan Gao, **Javier Fernandez-Marques**, Titouan Parcollet, Abhinav Mehrotra, and Nicholas D. Lane. *Federated Self-supervised Speech Representations: Are We There Yet*. Interspeech Conference (INTERSPEECH), 2022.
- ICLR 2022 Xinchu Qiu*, **Javier Fernandez-Marques***, Pedro Porto Buarque de Gusmao, Yan Gao, Titouan Parcollet, and Nicholas D. Lane. *ZeroFL: Efficient On-Device Training for Federated Learning with Local Sparsity*. International Conference on Learning Representations (ICLR), 2022. (*equal contrib.)
- FCCM 2021 Stylianos I. Venieris*, **Javier Fernandez-Marques***, Nicholas D. Lane. *unzipFPGA: Enhancing FPGA-based CNN Engines with On-the-Fly Weights Generation*. IEEE International Symposium on Field-Programmable Custom Computing Machines (FCCM), 2021. (*equal contribution)
- ICLR 2021 Shyam A. Tailor*, **Javier Fernandez-Marques***, Nicholas D. Lane. *Degree-Quant: Quantization-Aware Training for Graph Neural Networks*. International Conference on Learning Representations (ICLR), 2021. (*equal contribution)
- MLSys 2020 **Javier Fernandez-Marques**, Paul N. Whatmough, Andrew Mundy and Matthew Mattina. *Searching for Winograd-aware Quantized Networks*. Conference on Machine Learning Systems (MLSys), 2020
- ICLR 2019 Milad Alizadeh, **Javier Fernandez-Marques**, Nicholas D. Lane and Yarin Gal. *A Systematic Study of Binary Neural Networks' Optimisation*. International Conference on Learning Representations (ICLR), 2019
- IJCAI 2018 Vincent W.-S. Tseng, Sourav Bhattacharya, **Javier Fernandez-Marques**, Milad Alizadeh, Catherine Tong and Nicholas D. Lane. *Deterministic Binary Filters for Convolutional Neural Networks*. International Joint Conference on Artificial Intelligence (IJCAI), 2018
- ICIP 2015 Marting Maška, Cristina Ederra, **Javier Fernandez-Marques**, Arrate Muñoz-Barrutia, Michal Kozubek and Carlos Ortiz-de-Solórzano. *Quantification of the 3D Collagen Network Geometry in Confocal Reflection Microscopy*. International Conference on Image Processing (ICIP), 2015